

locosim ROS2 Humble — Ubuntu 22.04 Installation

This workspace is the ROS2 **Humble** port of locosim, targeting **Ubuntu 22.04 native** and **Gazebo Classic (gazebo11)** — not Ignition / Gazebo Sim.

Prerequisites (apt)

```
sudo apt update && sudo apt install -y \  
  ros-humble-desktop \  
  ros-humble-ros2-control \  
  ros-humble-ros2-controllers \  
  ros-humble-gazebo-ros2-control \  
  ros-humble-gazebo-ros-pkgs \  
  ros-humble-gazebo-plugins \  
  ros-humble-xacro \  
  ros-humble-robot-state-publisher \  
  ros-humble-joint-state-publisher \  
  ros-humble-joint-state-publisher-gui \  
  ros-humble-rviz2 \  
  ros-humble-tf2-ros \  
  ros-humble-tf2-tools \  
  ros-humble-rqt* \  
  python3-colcon-common-extensions \  
  python3-rosdep \  
  python3-vcstool \  
  libpinocchio-dev \  
  python3-pin
```

`srdfdom` is required by `base_hardware_interface` ; install it with
`sudo apt install -y ros-humble-srdfdom` if it is not already pulled in.

Python deps (pip)

```
pip3 install numpy scipy matplotlib
```

Build

```
mkdir -p ~/ros2_ws/src  
cd ~/ros2_ws/src  
# copy or clone locosim_ros2 here, e.g.:  
# cp -r /path/to/locosim_ros2/* ~/ros2_ws/src/  
cd ~/ros2_ws  
colcon build --symlink-install  
source install/setup.bash
```

Run (Go1 reference robot, Gazebo Classic)

```
# terminal 1 – bring up Gazebo Classic + spawn Go1 + start ros2_control controllers
ros2 launch ros_impedance_controller ros_impedance_controller.launch.py robot_name:=go1

# (visualization only – RViz + robot_state_publisher, no Gazebo)
ros2 launch ros_impedance_controller visualize.launch.py robot_name:=go1
```

Notes

- Source your ROS2 environment (`source /opt/ros/humble/setup.bash`) before building.
- Run `rosdep install --from-paths src --ignore-src -r -y` from `~/ros2_ws` to resolve any remaining package dependencies automatically.
- Gazebo Classic is used on purpose (`gazebo_ros` , `libgazebo_ros2_control.so` , `libgazebo_ros_init.so` , `libgazebo_ros_factory.so`). Do **not** install the Ignition/ `gz` stack for this workspace.